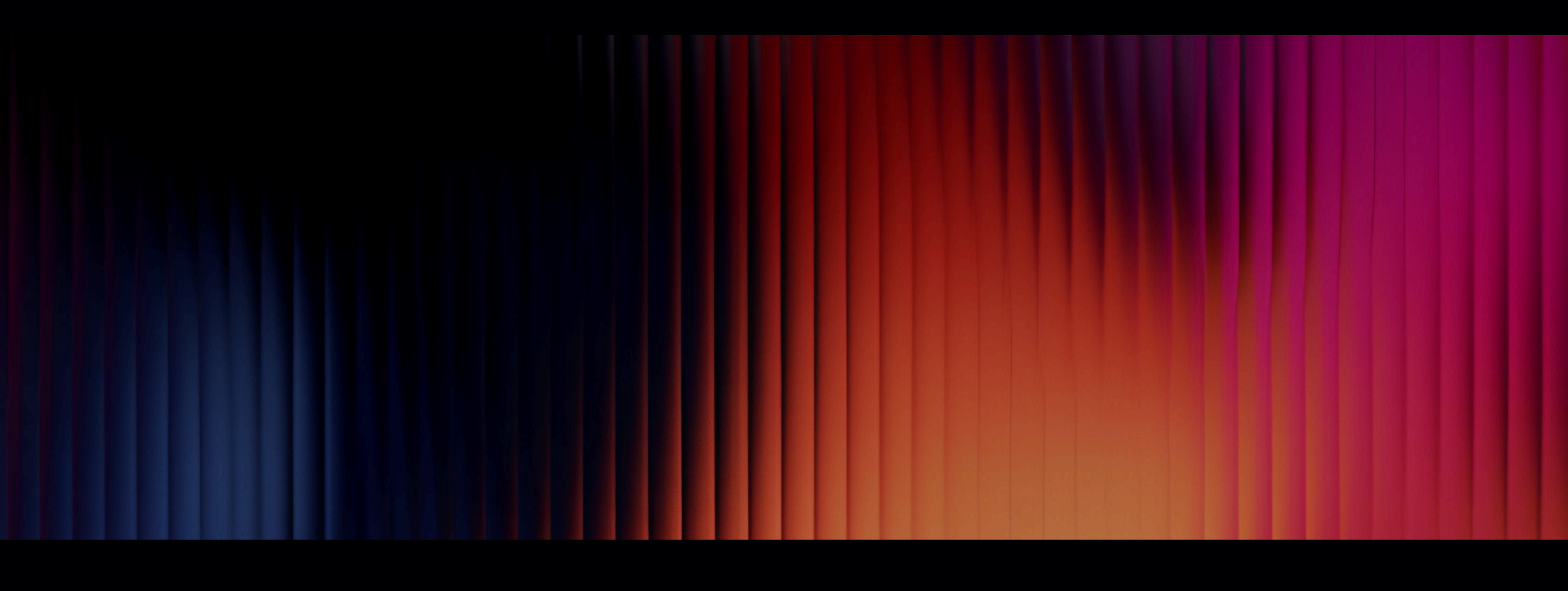
PulseBlend

A new way to experience Music & Colours

TANGIBLE
INTERACTION DESIGN

ADITI CHAWARE
ANOUSHKA D.
RIYA PHADNIS
SHREYAS
TANISHKA M.
YASHVI PAREKH



Project Overview

PulseBlend is a premium tangible interaction installation that transforms the physiological **rhythms of two people** into a shared **musical experience**. Users select the language of the experience, place their fingers on heart-rate sensors and allow the system to measure their heartbeats. Their individual rhythms are visualized, blended into a shared pulse, and used to select a song with a corresponding BPM. The experience combines sensing, physical interaction, music, light, motion and visual storytelling.

Target Audience

The primary audience is young adults and pairs aged **18–30,** comfortable with experimental technology and interested in music, art, design, immersive experiences, and discovering new forms of social interaction

The secondary audience includes friends, siblings, exhibition visitors, museum and gallery audiences, and music and technology enthusiasts.

Target Space

PulseBlend is designed as a third-space experience such as cafés, lounges, cultural spaces, community hubs, music spaces, and lifestyle venues, where two people can pause, interact, and create a shared musical experience.

Cafes

Art Installation

Experience spaces

Project Goal

Create Connection	Make the invisible visible	Reduce UI Interaction	Build Anticipation Through Play
Encourage two people to participate simultaneously to be a part of the experience	Heartbeats are normally invisible physiological signals. PulseBlend visualizes them through, music, light and motion.	Eliminate menus and touchscreen interaction. Let users interact naturally through physical controls and heart-rate sensors.	Gradually move users from curiosity to analysis to synchronization to musical reveal, while making technology feel like a fun.

Design Process

A cycle of research, ideation, prototyping, testing, and refinement.

Discover	Define	Ideate	Prototype	Build	Refine
Explored the relationship between music, emotion, heart rate, shared experiences, and tangible interaction.	Identified the opportunity to use two heartbeats as a shared input for music.	Explored concepts around rhythm, synchronization, music matching, physical controls, light, and visualization.	Tested physical forms, sensors, displays, LEDs, BPM-to-song mapping, circuits, and software.	Integrated the enclosure, sensors, controls, screen, electronics, LEDs, and software into the final experience.	Simplified the interaction to make it feel calm, intuitive, and emotionally engaging.

Research

1. How we started

Transitioned from broad human-technology exploration to focusing on music as a medium to connect two people through physical and physiological interaction, bypassing standard screen interfaces.

2. Physiological Input

Selected heart rate as the core continuous input, aiming to transform biometric data from raw numerical metrics into an experiential parameter users can hear, see, and feel.

3. Musical Translation

Bridged physiological rhythm to music via BPM, utilizing the heartbeat as an active, dynamic parameter to drive an evolving musical system.

4. Tangible & Social Interaction

Designed for tangible, two-person collaboration, ensuring the users' physical bodies act as the interface and their mutual relationship becomes the core interaction.

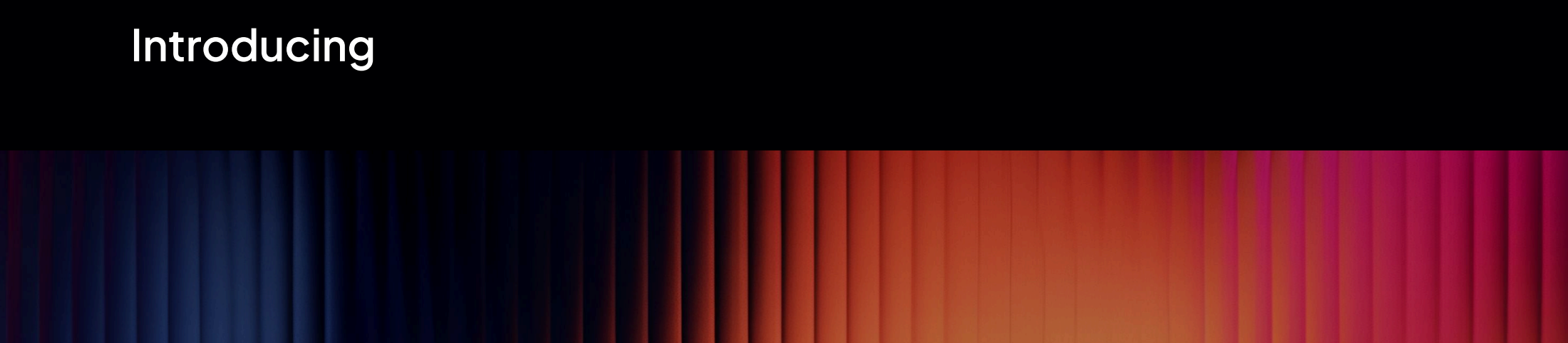
5. Guiding Design Principles

Established four foundational tenets: feel instead of measure, interact instead of operate, participate together rather than individually, and reveal the experience gradually through sensory feedback.

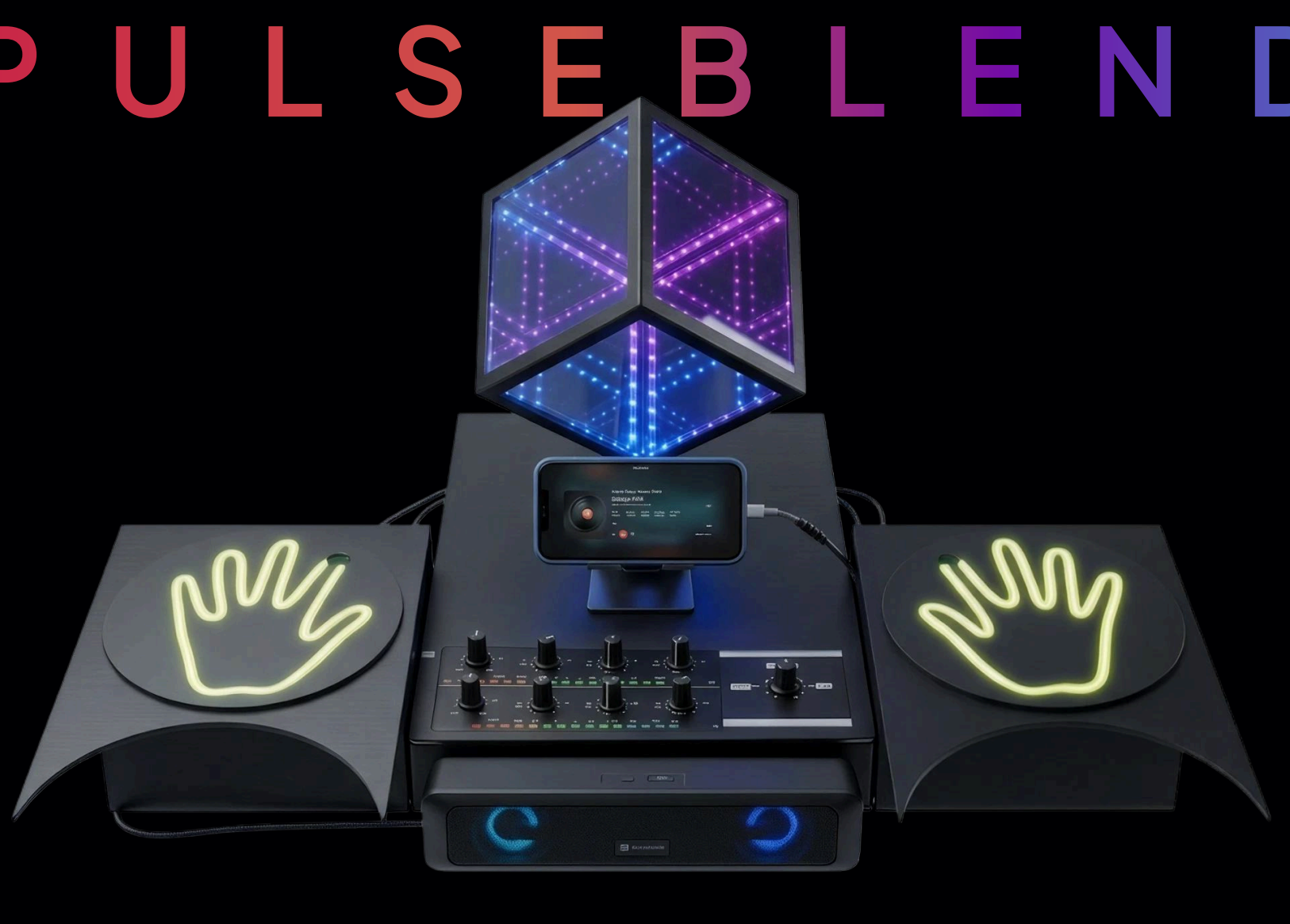
6. Concept Synthesis

Synthesized these insights into PulseBlend: a tangible audiovisual installation that merges two participants' physiological rhythms and musical identities into a unified, shared experience.

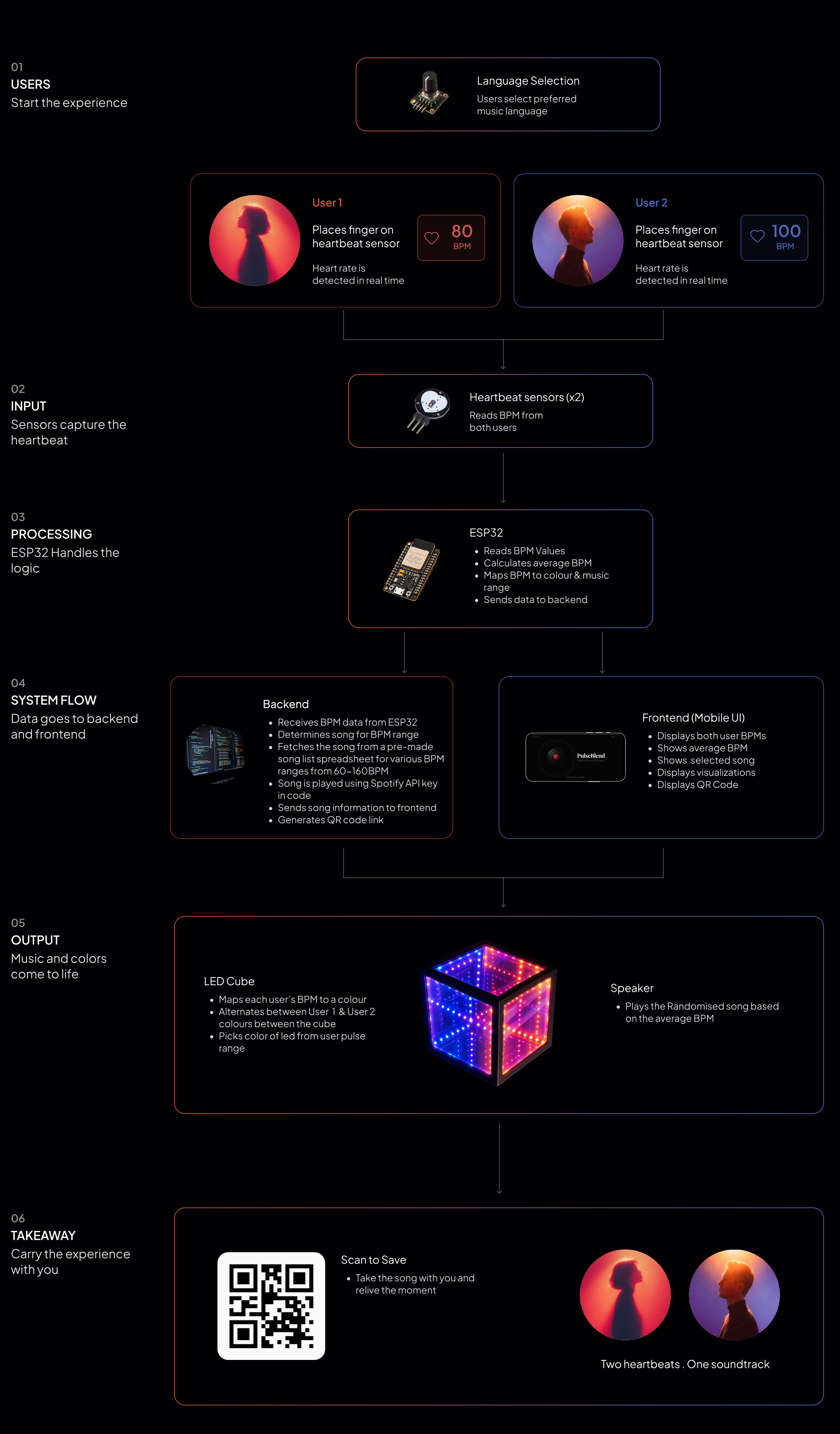
Introducing



P U L S E B L E N D



Experience Flow



01
USERS
Start the experience

Language Selection
Users select preferred music language

User 1
Places finger on heartbeat sensor
80 BPM
Heart rate is detected in real time

User 2
Places finger on heartbeat sensor
100 BPM
Heart rate is detected in real time

02
INPUT
Sensors capture the heartbeat

Heartbeat sensors (x2)
Reads BPM from both users

03
PROCESSING
ESP32 Handles the logic

ESP32
- Reads BPM Values
- Calculates average BPM
- Maps BPM to colour & music range
- Sends data to backend

04
SYSTEM FLOW
Data goes to backend and frontend

Backend
- Receives BPM data from ESP32
- Determines song for BPM range
- Fetches the song from a pre-made song list spreadsheet for various BPM ranges from 60–160BPM
- Song is played using Spotify API key in code
- Sends song information to frontend
- Generates QR code link

Frontend (Mobile UI)
- Displays both user BPMs
- Shows average BPM
- Shows selected song
- Displays visualizations
- Displays QR Code

05
OUTPUT
Music and colors come to life

LED Cube
- Maps each user's BPM to a colour
- Alternates between User 1 & User 2 colours between the cube
- Picks color of led from user pulse range

Speaker
- Plays the Randomised song based on the average BPM

06
TAKEAWAY
Carry the experience with you

Scan to Save
- Take the song with you and relive the moment

Two heartbeats . One soundtrack

Components Used

Pulse Blend integrates physical sensors, embedded electronics, and a connected mobile interface to transform two users' heartbeats into a shared musical experience. The ESP32 acts as the central controller, processing BPM inputs and coordinating the cube's lighting, physical controls, and digital music interaction.

Arduino ESP32

Main microcontroller for sensor input, BPM processing, and system control.

Pulse Beat Sensor x2

Capture heart rate data from two users.

Potentiometer

Rotational input for Language Selection

Various Wires

Connect sensors, controls, LEDs, and controller.

On off switch

Switches of entire System

LED Strip wire

Codable Color Changeable Wire with individual

External Speaker System

Plays selected muisc

USBC Cable

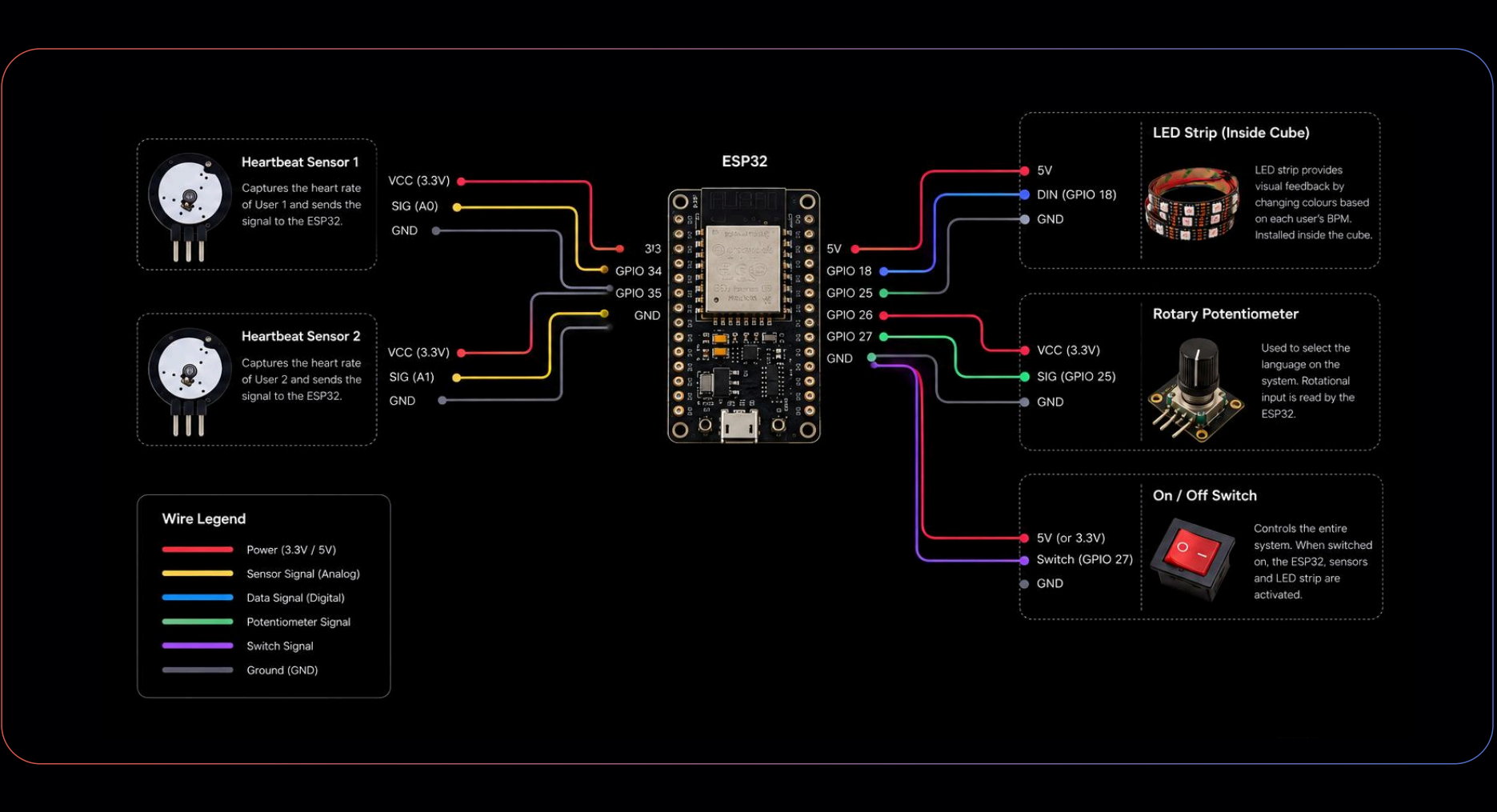
Powers or programs the microcontroller.

Mobile User Interface

Controls and shows users the current states

Circuit Diagram and Wiring

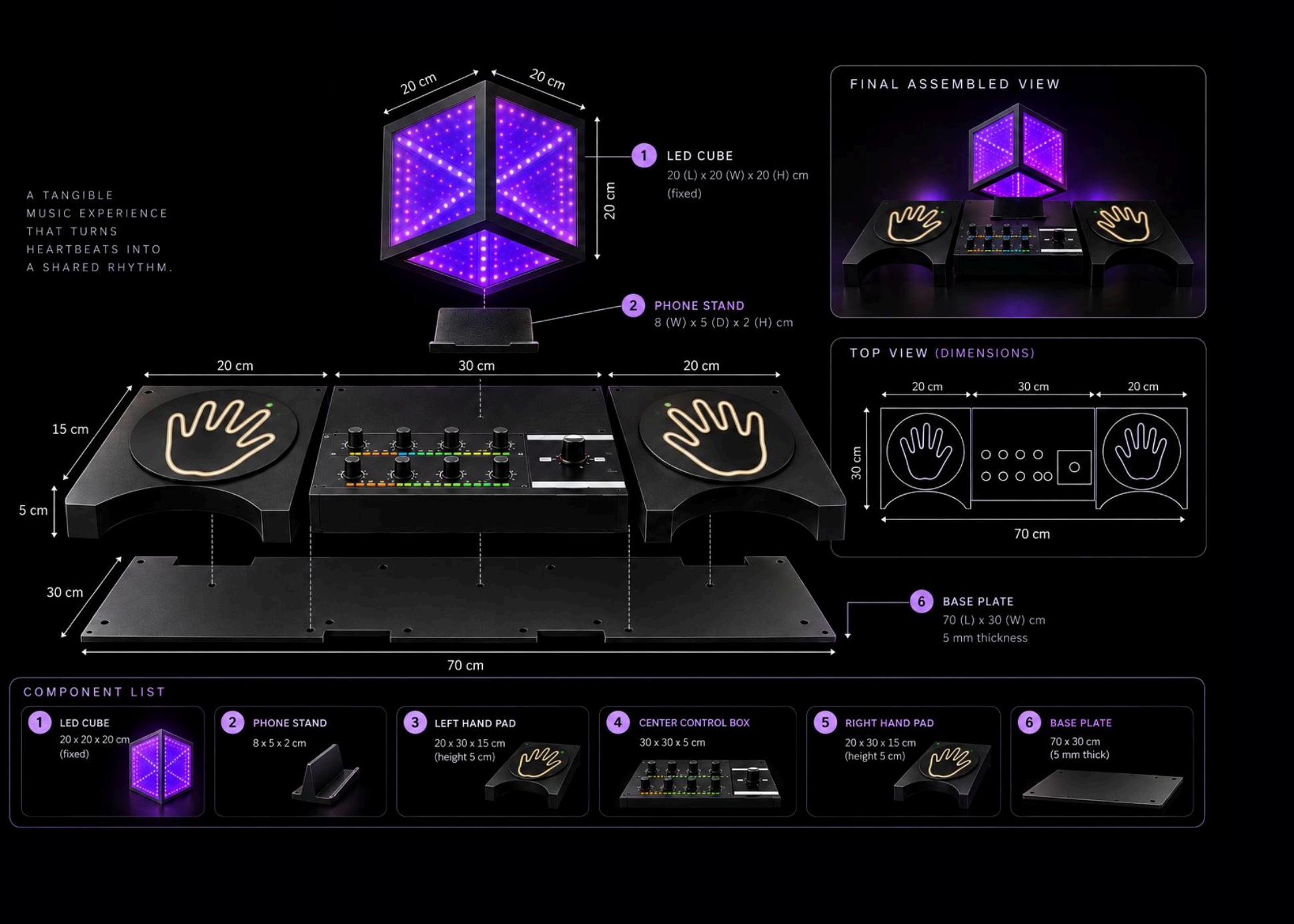
The circuit integrates both heartbeat sensors, the ESP32 microcontroller, LED strips, and physical input controls through wired connections. The ESP32 receives individual BPM readings, calculates the combined average, and coordinates the corresponding lighting response. The potentiometer enables language selection, while the on/off switch controls system activation. Music playback is handled through an external speaker system, with the mobile interface supporting the connected digital experience.



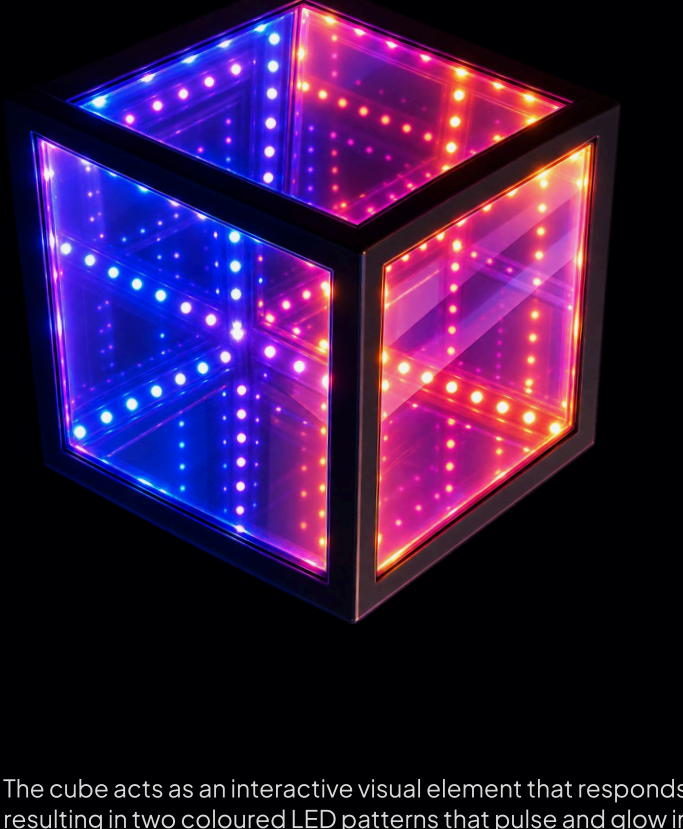
Heartbeat Sensor 1
Captures the heart rate of User 1 and sends the signal to the ESP32.
VCC (3.3V)
SIG (A0)
GND
Heartbeat Sensor 2
Captures the heart rate of User 2 and sends the signal to the ESP32.
VCC (3.3V)
SIG (A1)
GND
ESP32
3.3
GPIO 34
GPIO 35
GND
5V
GPIO 18
GPIO 25
GPIO 26
GPIO 27
GND
LED Strip (Inside Cube)
5V
DIN (GPIO 18)
GND
LED strip provides visual feedback by changing colours based on each user's BPM. Installed inside the cube.
Rotary Potentiometer
VCC (3.3V)
SIG (GPIO 25)
GND
Used to select the language on the system. Rotational input is read by the ESP32.
On / Off Switch
5V (or 3.3V)
Switch (GPIO 27)
GND
Controls the entire system. When switched on, the ESP32, sensors and LED strip are activated.
Wire Legend
Power (3.3V / 5V)
Sensor Signal (Analog)
Data Signal (Digital)
Potentiometer Signal
Switch Signal
Ground (GND)

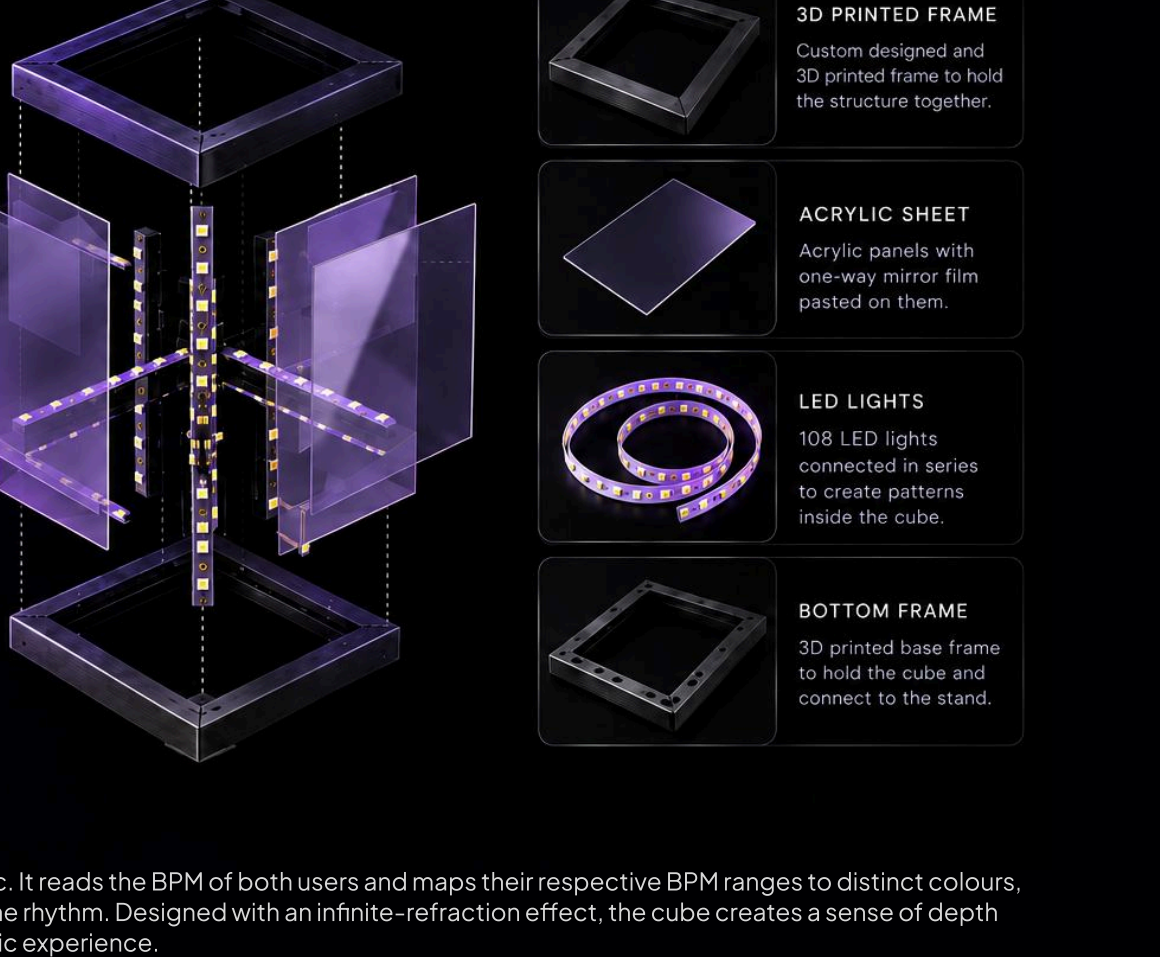
The ESP32 serves as the bridge between physical interaction and digital music selection, enabling Pulse Blend to translate physiological inputs into a responsive, shared experience.

Physical Components



A TANGIBLE MUSIC EXPERIENCE THAT TURNS HEARTBEATS INTO A SHARED RHYTHM.
20 cm
20 cm
20 cm
1 LED CUBE
20 (L) x 20 (W) x 20 (H) cm
(fixed)
2 PHONE STAND
8 (W) x 5 (D) x 2 (H) cm
20 cm
30 cm
20 cm
15 cm
5 cm
30 cm
70 cm
6 BASE PLATE
70 (L) x 30 (W) cm
5 mm thickness
FINAL ASSEMBLED VIEW
TOP VIEW (DIMENSIONS)
20 cm
30 cm
20 cm
30 cm
70 cm
COMPONENT LIST
1 LED CUBE
20 x 20 x 20 cm
(fixed)
2 PHONE STAND
8 x 5 x 2 cm
3 LEFT HAND PAD
20 x 30 x 15 cm
(height 5 cm)
4 CENTER CONTROL BOX
30 x 30 x 5 cm
5 RIGHT HAND PAD
20 x 30 x 15 cm
(height 5 cm)
6 BASE PLATE
70 x 30 cm
(5 mm thick)

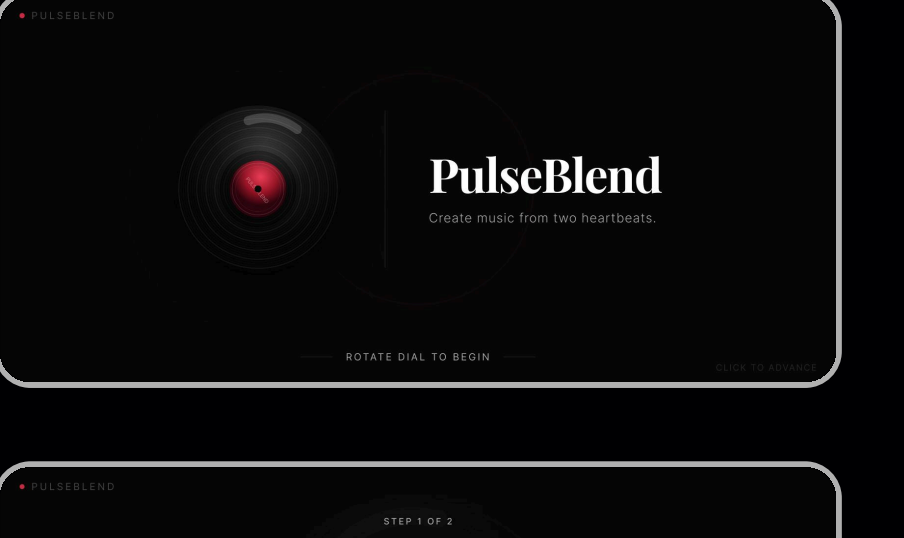




3D PRINTED FRAME
Custom designed and 3D printed frame to hold the structure together.
ACRYLIC SHEET
Acrylic panels with one-way mirror film pasted on them.
LED LIGHTS
108 LED lights connected in series to create patterns inside the cube.
BOTTOM FRAME
3D printed base frame to hold the cube and connect to the stand.

The cube acts as an interactive visual element that responds to the music. It reads the BPM of both users and maps their respective BPM ranges to distinct colours, resulting in two coloured LED patterns that pulse and glow in sync with the rhythm. Designed with an infinite-refraction effect, the cube creates a sense of depth and continuous light, adding a visually immersive layer to the overall music experience.

React Front End UI

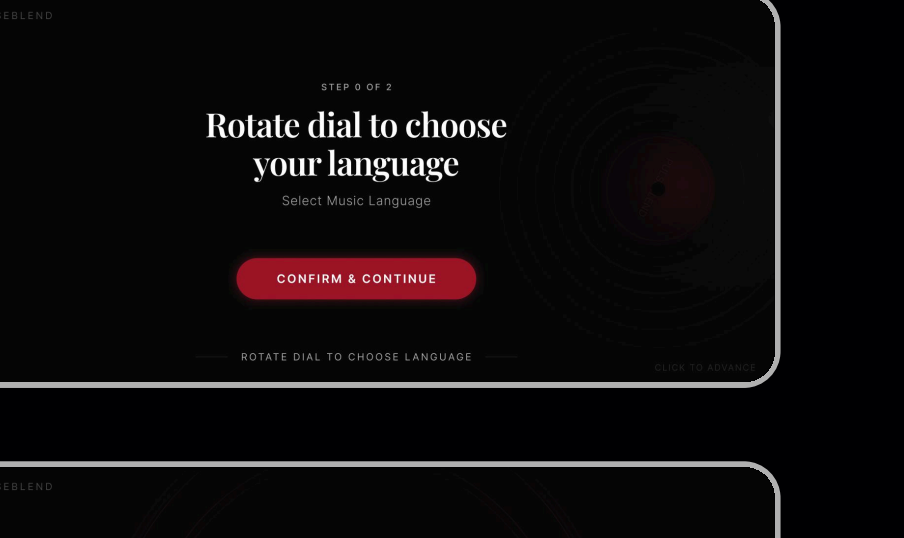


PULSEBLEND

PulseBlend

Create music from two heartbeats.

ROTATE DIAL TO BEGIN



PULSEBLEND

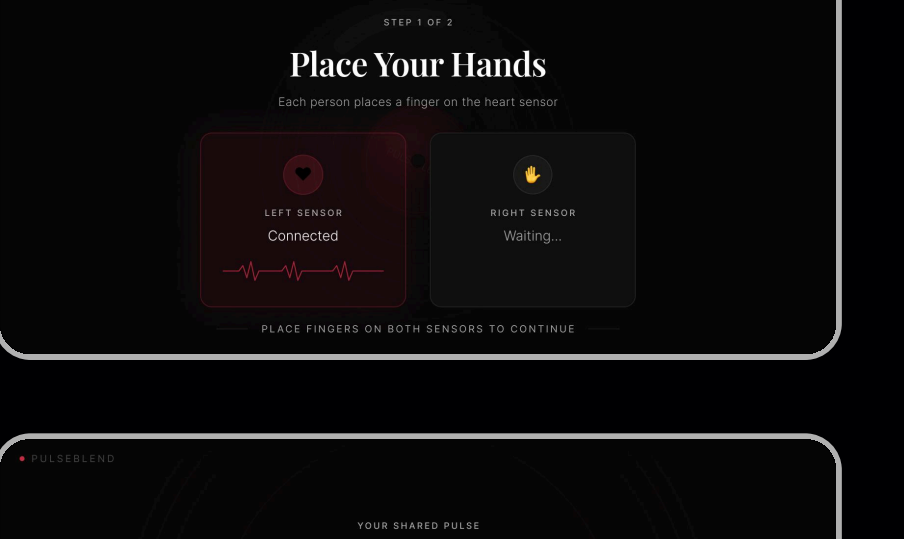
STEP 0 OF 2

Rotate dial to choose your language

Select Music Language

CONFIRM & CONTINUE

ROTATE DIAL TO CHOOSE LANGUAGE



PULSEBLEND

STEP 1 OF 2

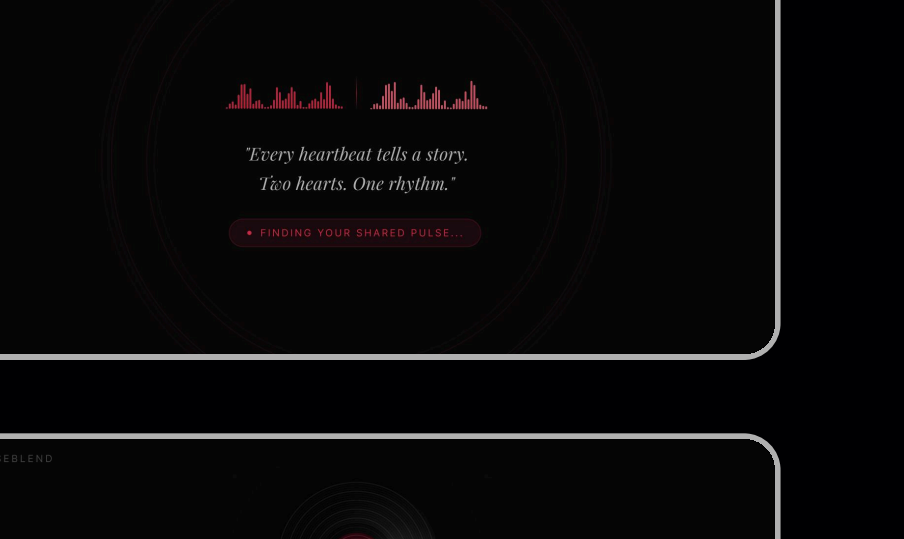
Place Your Hands

Each person places a finger on the heart sensor

LEFT SENSOR
Connected

RIGHT SENSOR
Waiting...

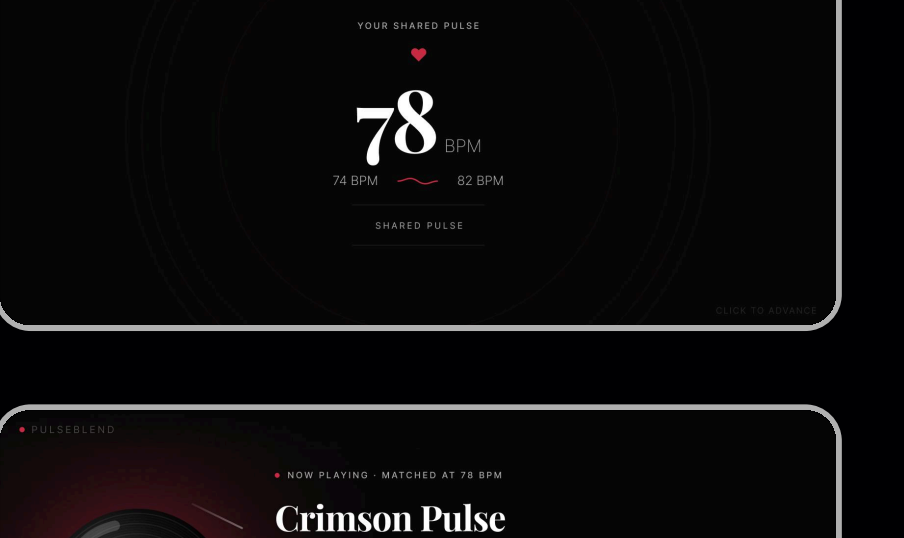
PLACE FINGERS ON BOTH SENSORS TO CONTINUE



PULSEBLEND

"Every heartbeat tells a story.
Two hearts. One rhythm."

FINDING YOUR SHARED PULSE...



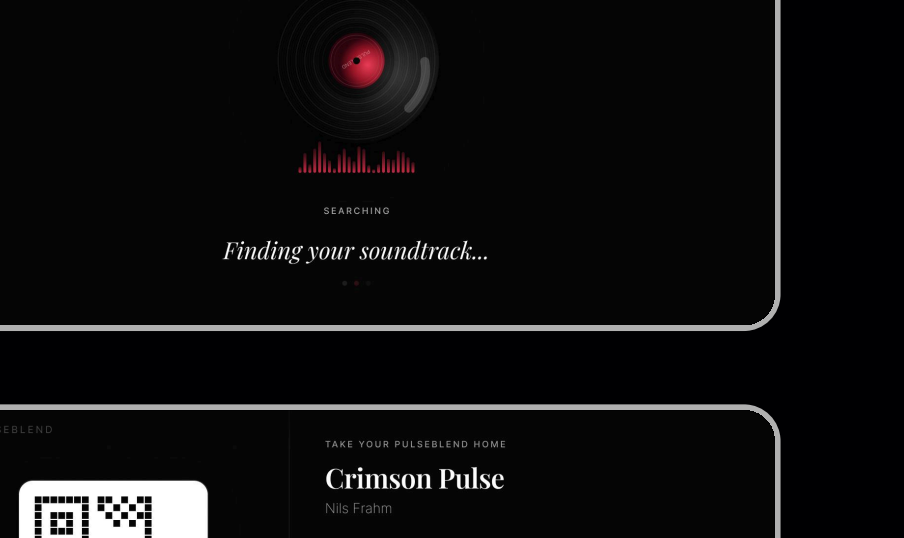
PULSEBLEND

YOUR SHARED PULSE

78 BPM

74 BPM 82 BPM

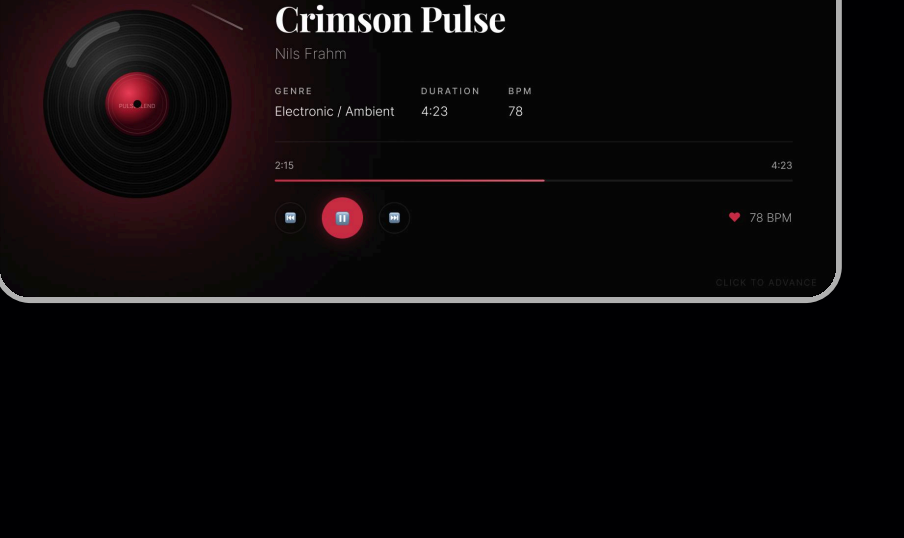
SHARED PULSE



PULSEBLEND

SEARCHING

Finding your soundtrack...



PULSEBLEND

NOW PLAYING · MATCHED AT 78 BPM

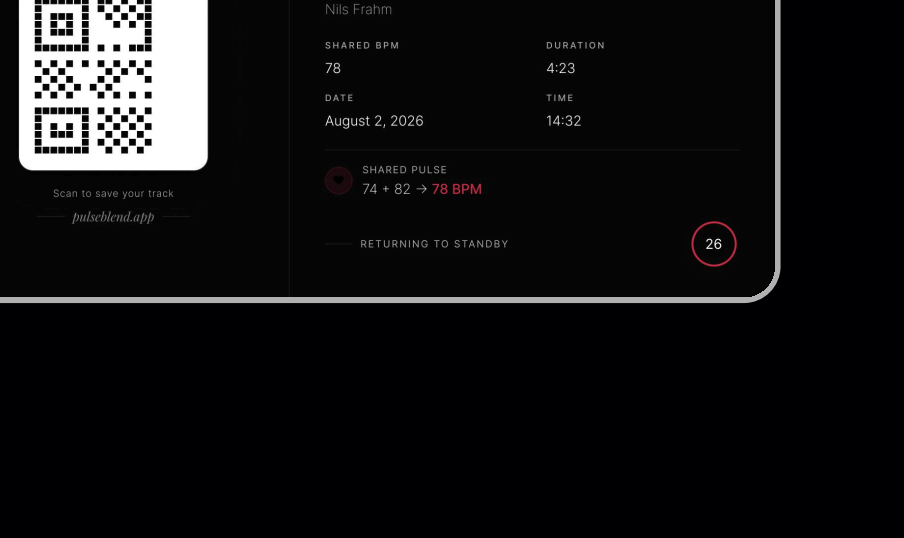
Crimson Pulse

Nils Frahm

GENRE	DURATION	BPM
Electronic / Ambient	4:23	78

2:15 4:23

78 BPM



PULSEBLEND

Scan to save your track

pulseblend.app

TAKE YOUR PULSEBLEND HOME

Crimson Pulse

Nils Frahm

SHARED BPM	DURATION
78	4:23
DATE	TIME
August 2, 2026	14:32

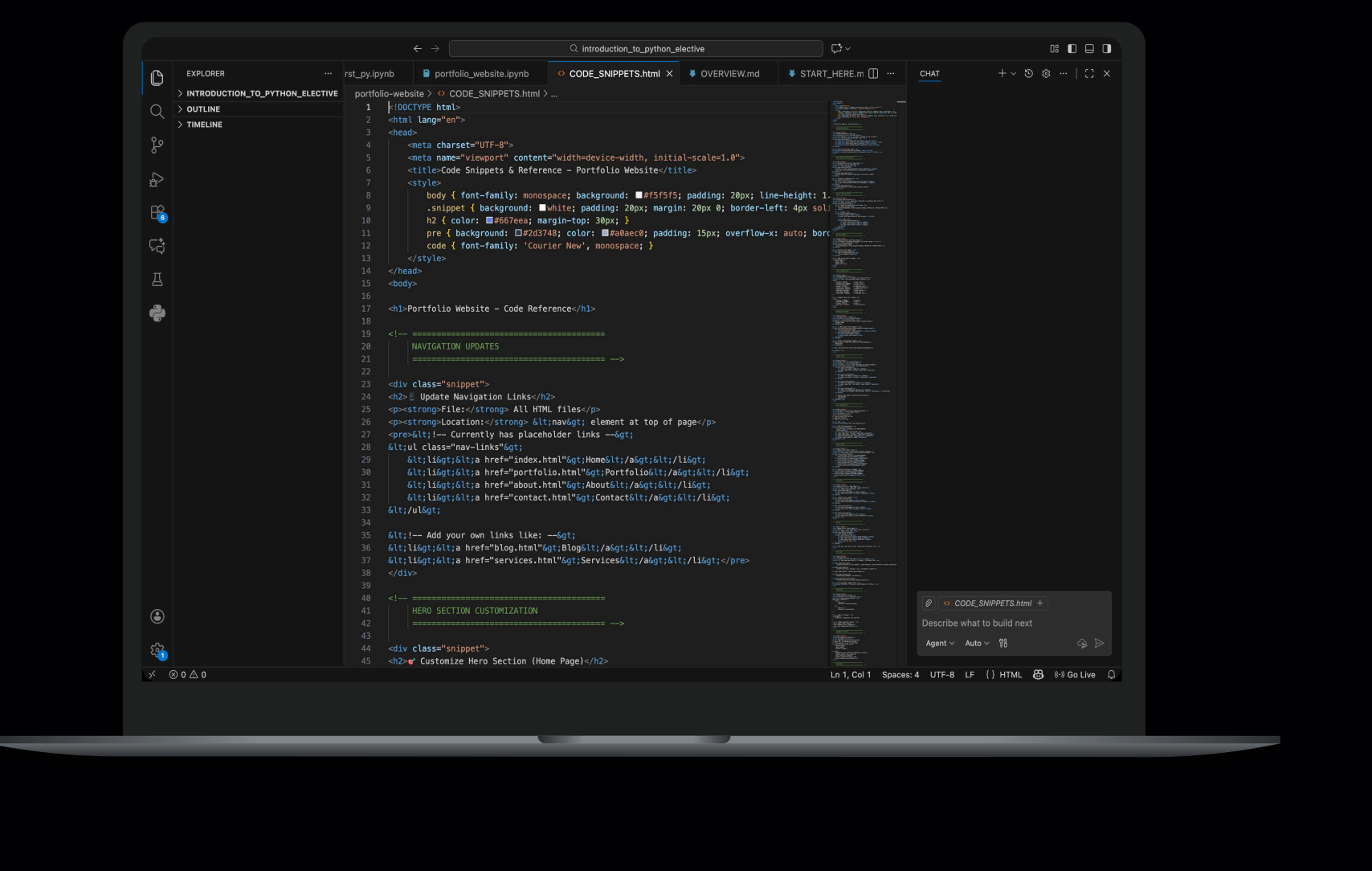
SHARED PULSE
74 + 82 → 78 BPM

RETURNING TO STANDBY

26

Coding the backend

The Pulse Blend backend was Vibecoded in Python using VS Code with Claude Code, with Flask managing communication between the ESP32, Spotify, and the React-based mobile interface. The system processes incoming BPM data, filters a CSV-based song library by average heart rate and language, and triggers Spotify playback. It also serves live session information to the frontend, connecting the physical interaction with the digital music experience.



introduction_to_python_elective
EXPLORER
INTRODUCTION_TO_PYTHON_ELECTIVE
OUTLINE
TIMELINE
rst_py.ipynb
portfolio_website.ipynb
CODE_SNIPPETS.html
OVERVIEW.md
START_HERE.m
CHAT
portfolio-website > CODE_SNIPPETS.html > ...
Describe what to build next
Agent
Auto
Ln 1, Col 1
Spaces: 4
UTF-8
LF
HTML
Go Live

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        code { font-family: 'Courier New', monospace; }
    </style>
</head>
<body>

<h1>Portfolio Website - Code Reference</h1>

<!-- ========================================
     NAVIGATION UPDATES
     ======================================== -->

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<p><strong>Location:</strong> &lt;nav&gt; element at top of page</p>
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    &lt;li&gt;&lt;a href="portfolio.html"&gt;Portfolio&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="about.html"&gt;About&lt;/a&gt;&lt;/li&gt;
    &lt;li&gt;&lt;a href="contact.html"&gt;Contact&lt;/a&gt;&lt;/li&gt;
&lt;/ul&gt;

&lt;!-- Add your own links like: --&gt;
&lt;li&gt;&lt;a href="blog.html"&gt;Blog&lt;/a&gt;&lt;/li&gt;
&lt;li&gt;&lt;a href="services.html"&gt;Services&lt;/a&gt;&lt;/li&gt;</pre>
</div>

<!-- ========================================
     HERO SECTION CUSTOMIZATION
     ======================================== -->

<div class="snippet">
<h2> Customize Hero Section (Home Page)</h2>
```

1. Serial Data Handling

The Python backend, developed in VS Code, reads the ESP32's serial output on COM6 at 115200 baud. It extracts both users' BPM values and identifies the final shared BPM for the music selection process.

2. CSV-Based BPM & Colour Mapping

Using Pandas, the backend reads CSV spreadsheets containing song data and BPM-to-colour mappings. The song spreadsheet stores track information, BPM ranges, and language categories, while the colour spreadsheet maps individual BPM ranges to corresponding LED colours. This enables the system to identify suitable songs and colour responses based on the users' heart rates.

3. Randomised Song Selection

The backend filters songs.csv using the average BPM range and selected language, then randomly selects a song from the matching entries. For example, an average BPM of 92 narrows the selection to the 90–95 BPM range. The random selection adds variety, allowing users with similar heart rates to experience different songs.

4. Spotify Integration

Spotipy and Spotify OAuth handle authentication, track search, and playback commands. Once a song is randomly selected, the backend retrieves its Spotify URI, album artwork, song URL, and duration, then triggers playback through Spotify.

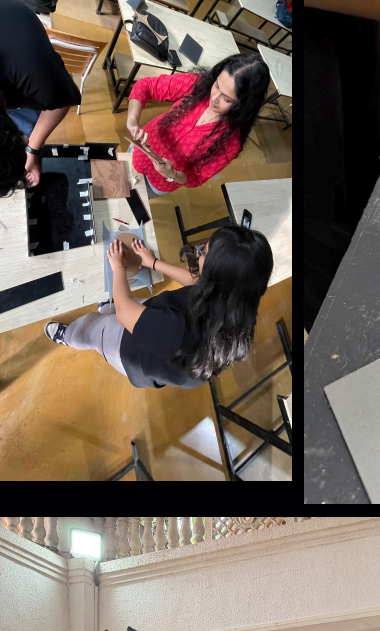
5. Frontend Communication

Flask exposes API endpoints such as /status, /language, /play, /playpause, and /next. The React interface fetches session information from the backend to display BPM readings, selected song details, and playback controls

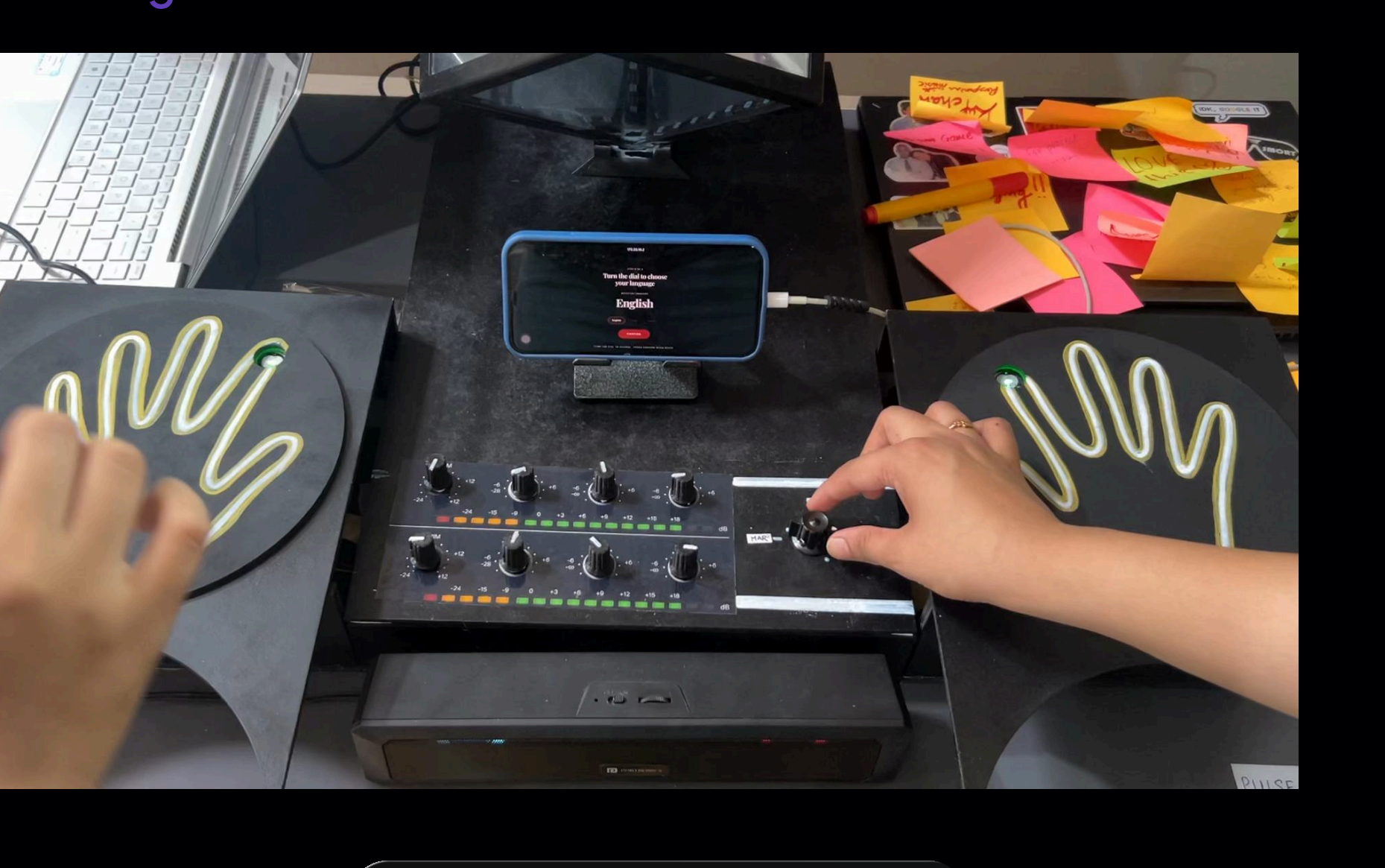
6. LED Synchronisation & QR Takeaway

The backend sends the selected song's BPM back to the ESP32 over serial. The ESP32 uses the BPM-to-colour mapping to coordinate the LED response, alternating between User 1 and User 2's individual BPM colours. Meanwhile, the frontend receives the Spotify URL and track metadata to display the selected song and generate a scannable QR code for users to take the music with them.

Process Photos

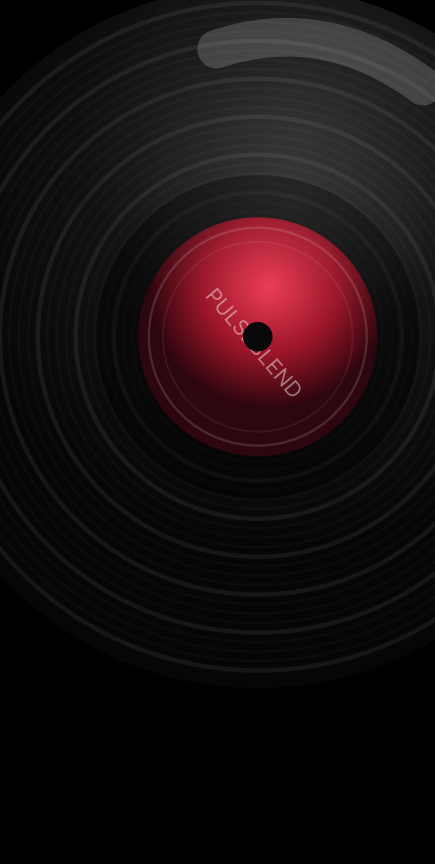


Working Video

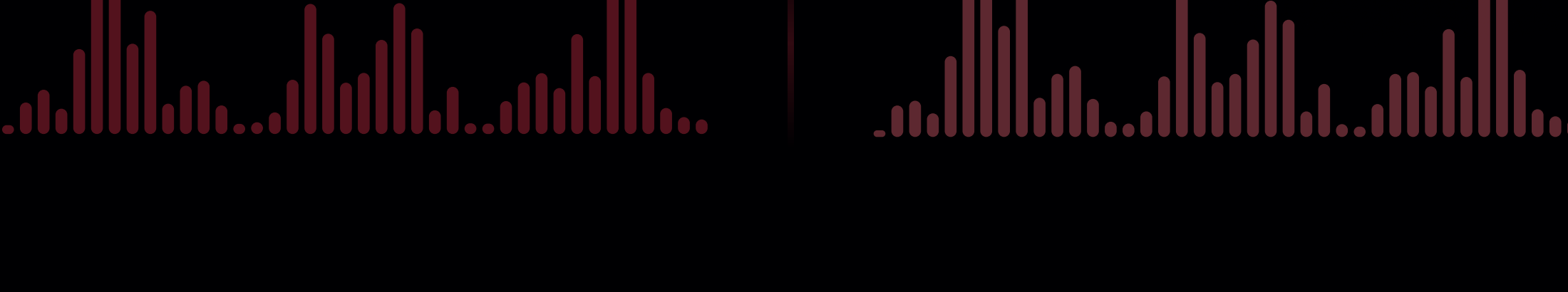


Link to working MODEL

Learnings



- Pulse Blend explores how tangible interaction and physical computing can transform physiological data into a shared social experience.
- By combining heartbeat sensing, BPM-based music selection, alternating LED feedback, physical controls, and a connected mobile interface, the project creates an interaction that exists across both physical and digital touchpoints.
- Designed for third spaces, Pulse Blend reimagines the role of interactive objects in social environments. Rather than treating heart rate as a standalone measurement, it uses the data as a playful input for music discovery, ambient interaction, and shared experiences.
- The final project brings together concept development, tangible product design, electronics prototyping, embedded programming, and UI/backend integration into one connected experience.



Thank You!